

Exercises for 12.1

• Sketch the level curves for $k = -2, -1, 0, 1, 2$

a) $z = \frac{1}{2}(x^2 + y^2)$

b) $z = x/y$

c) $z = x^2/y$

d) $z = x^2 + y$

e) $z = y - \sin x$

f) $w = x^2 + y^2 + z^2$

g) $w = e^{x^2 + y^2 + z^2}$